Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: SPV5_Kolkata_5MWp

Variant: Base case: 20deg tilt, no shading, 26x565W x 340 strings, 16x SG250HX

No 3D scene defined, no shadings

System power: 4995 kWp

Kolkata - India

Author

PVsyst V7.4.7
VC0, Simulation date:
28/10/26 12:20
with V7.4.7

Project summary

Geographical Site
Kolkata
India

Situation

Latitude	22.56 °N
Longitude	88.36 °E
Altitude	11 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Kolkata
Meteonorm 8.1 (1991-2012) - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation
Fixed plane
Tilt/Azimuth 20 / 0 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	8840 units
Pnom total	4995 kWp

Inverters

Nb. of units	16 units
Pnom total	3999 kWac
Pnom ratio	1.249

Results summary

Produced Energy	7041025 kWh/year	Specific production	1410 kWh/kWp/year	Perf. Ratio PR	85.20 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		Sheds configuration	Models used	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	20 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon
Free Horizon

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Waaree Energies
Model	BiN-08-565
(Custom parameters definition)	
Unit Nom. Power	565 Wp
Number of PV modules	8840 units
Nominal (STC)	4995 kWp
Modules	340 string x 26 In series
At operating cond. (50°C)	
Pmpp	4620 kWp
U mpp	1034 V
I mpp	4467 A

Inverter

Manufacturer	Sungrow
Model	SG250-HX
(Original PVsyst database)	
Unit Nom. Power	250 kWac
Number of inverters	192 * MPPT 8% 16 units
Total power	3999 kWac
Operating voltage	500-1450 V
Pnom ratio (DC:AC)	1.25
No power sharing between MPPTs	

Total PV power

Nominal (STC)	4995 kWp
Total	8840 modules
Module area	22836 m²

Total inverter power

Total power	3999 kWac
Number of inverters	16 units
Pnom ratio	1.25

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 3.8 mΩ
Loss Fraction 1.5 % at STC

Module Quality Loss
Loss Fraction -0.5 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

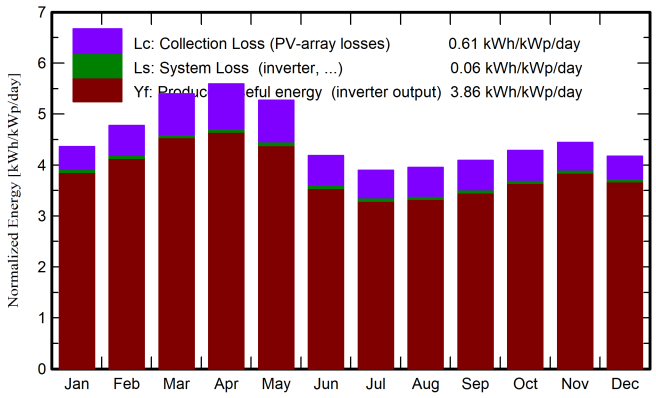
0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

Main results

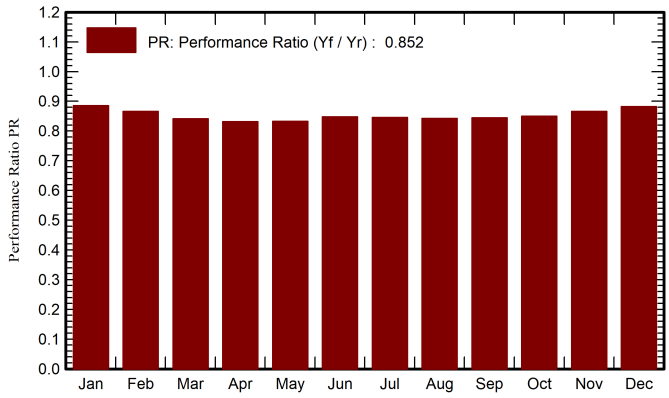
System Production

Produced Energy	7041025 kWh/year	Specific production	1410 kWh/kWp/year
		Perf. Ratio PR	85.20 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

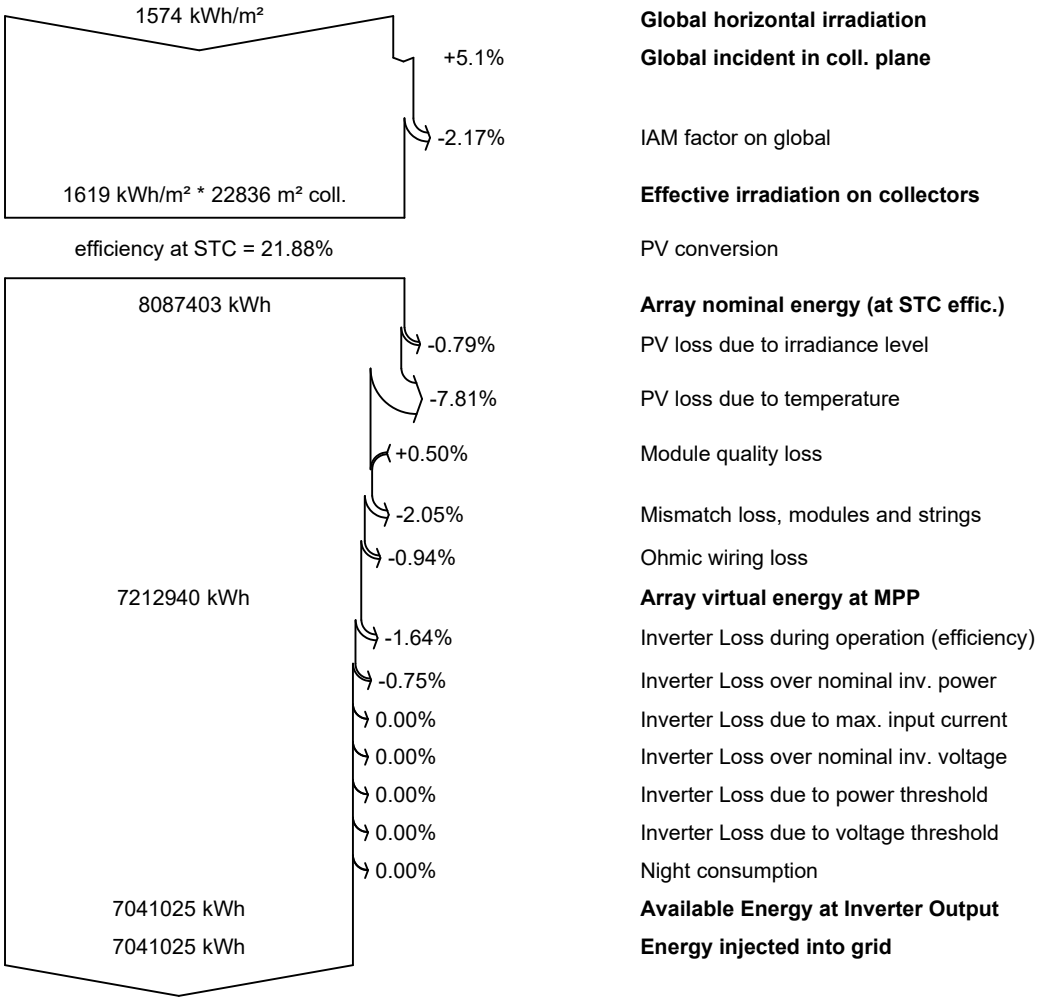
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	112.1	60.55	17.45	135.2	132.6	606782	597879	0.885
February	118.0	67.00	22.10	133.7	131.2	586675	577985	0.866
March	156.9	85.04	27.55	167.3	164.0	713552	703117	0.841
April	166.5	88.89	30.18	167.7	164.3	707580	696984	0.832
May	171.7	96.70	31.43	163.5	159.7	690669	679368	0.832
June	134.0	93.63	30.47	125.5	122.2	541743	531222	0.847
July	128.3	87.28	29.78	120.8	117.5	521107	510425	0.846
August	125.0	83.59	29.36	122.5	119.4	525901	515515	0.843
September	120.4	78.22	28.31	122.8	119.8	527506	518081	0.845
October	121.8	73.59	27.25	132.9	130.1	573696	564247	0.850
November	112.8	60.86	23.36	133.4	130.9	585549	576566	0.866
December	106.1	57.07	18.68	129.3	127.0	578689	569636	0.882
Year	1573.6	932.42	26.34	1654.6	1618.6	7159449	7041025	0.852

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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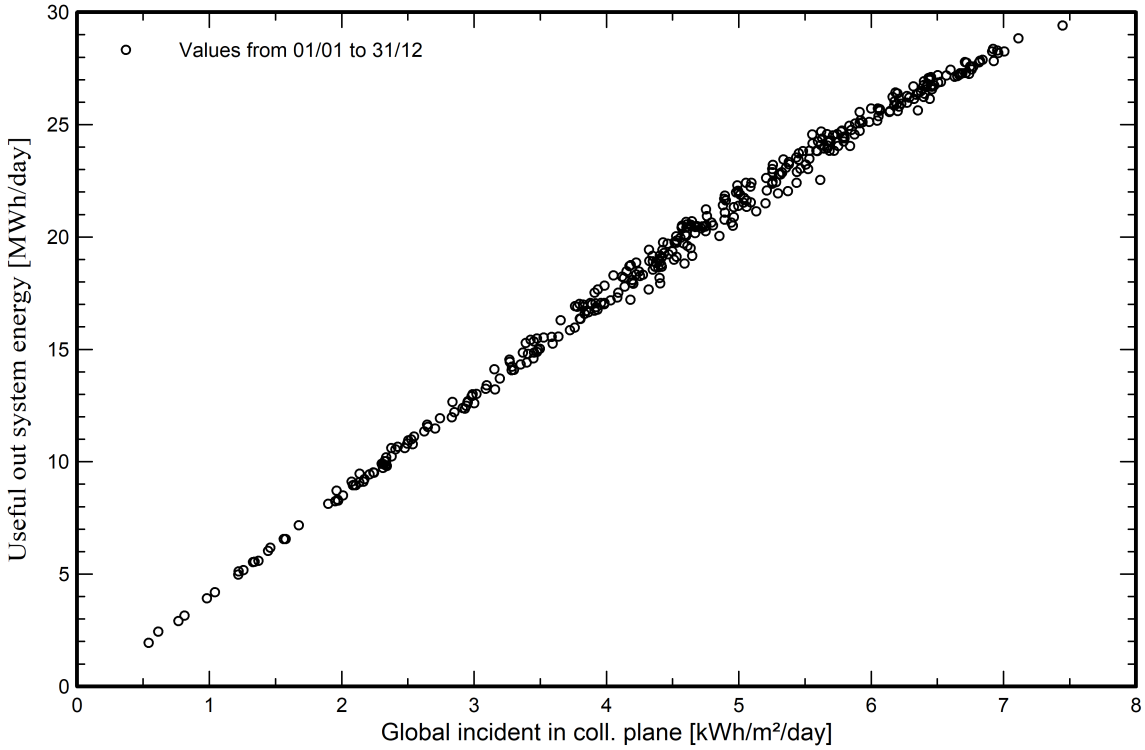
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

